

Multimodal Pedagogical Friction Detector: Fusing Facial Affect, Talk Moves, and LLM Reasoning to Surface Teachable Moments at Scale

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Abstract

We present an automated system for detecting *pedagogical friction*: moments where a low-quality instructional strategy co-occurs with elevated student confusion. Our pipeline fuses per-frame facial affect (DeepFace [1]), automatic speech recognition (Whisper [2]), and Talk Move classification (ELECTRA [3]) on a 30-second temporal grid, then applies `claude-sonnet-4-6` as a second-pass precision filter that supplies both a friction verdict and an actionable alternative strategy. Evaluated on a multimodal run on 10 min of TIMSS video, transcript-only analysis of **53 lessons** across 7 countries (5 113 bins), and human annotation on AU1, the pipeline achieves **F1 = 1.00** on that lesson (vs. $F1 = 0.10$ for the heuristic), reduces transcript-only candidates $7.6\times$ to 211 friction moments, and reveals a cross-method correlation of $r = -0.35$, marking annotation methodology as a first-class design variable.

1 Introduction

High-quality teaching requires *professional vision*, the ability to notice and interpret significant events in classroom interaction [4]. When student understanding stalls, the teacher’s next move is critical: giving the answer directly, pressing for rote recall, or ignoring the confusion all prevent the productive cognitive engagement necessary for deep learning [5]. We call such moments *pedagogical friction*: the co-occurrence of elevated student confusion and a low-quality instructional response.

Existing automated tools address only one modality at a time. Edu-ConvoKit [6] and Tutor CoPilot [7] analyse transcript structure but ignore student affect; facial-affect systems detect confusion but cannot interpret whether the teacher’s response was appropriate. The TIMSS 1999

Video Study [8] manually coded thousands of clips across seven countries, a gold standard requiring years of expert effort that cannot scale to real-time feedback.

We bridge this gap with a system that fuses vision and language signals on a shared 30-second temporal grid, applies `claude-sonnet-4-6` as a multimodal reranker generating concrete alternative strategies, and scales to 53 transcript lessons plus 54 TIMSS videos. On AU1, the two-stage design raises friction-detection F1 from 0.10 to 1.00; across the transcript corpus, it reduces candidate load by $7.6\times$. Code is available at <https://github.com/jw310-wjr/comp646-friction-detector>.

2 Related Work

Suresh et al. [3] fine-tune ELECTRA on Talk Move classification; Wang and Demszky [6] and Wang et al. [7] provide NLP utilities and an LLM copilot for educational dialogues—neither fuses discourse quality with student affect. D’Mello and Graesser [5] show confusion and frustration are most predictive of learning disruption, motivating Eq. 1. Hiebert et al. [8] and Sherin and van Es [4] demonstrate the value of systematic video coding and professional vision training, which our pipeline automates at scale.

3 Methodology

Figure 1 shows the two-stream architecture. A **Vision Stream** (blue) and a **Language Stream** (purple) run in parallel on a shared 30-second temporal grid, converge at a z-score filter, and produce a structured friction verdict via `claude-sonnet-4-6` fusion. Figure 2 shows a concrete input/output example for the highest-confidence friction moment in our dataset.

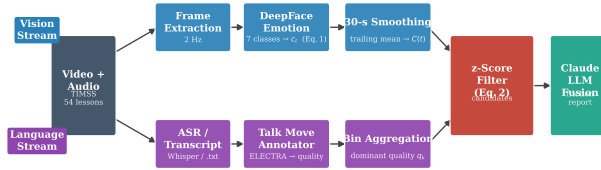


Figure 1: Two-stream pipeline: Vision (blue, Eq. 1) and Language (purple) streams share a 30-s grid; a z -score filter (Eq. 2) selects candidates for `claude-sonnet-4-6` friction verdict and alternative strategy.

3.1 Vision Stream: Confusion Timeline

Video is sampled at $r=2$ Hz. Each frame is passed to DeepFace [1], which returns a probability vector $\mathbf{p}_t \in \mathbb{R}^7$ over Ekman’s seven basic emotions. We map this to a scalar *confusion proxy* $c_t \in [0, 1]$:

$$c_t = \sum_{e \in \mathcal{E}} w_e \cdot p_t(e), \quad (1)$$

where weights are derived from the affect-outcome correlations in D’Mello and Graesser [5]: $w_{\text{fear}}=0.90$, $w_{\text{sad}}=0.85$, $w_{\text{disgust}}=0.50$, $w_{\text{angry}}=0.45$, $w_{\text{surprise}}=0.25$, $w_{\text{happy}}=w_{\text{neutral}}=0$ (higher weight = stronger empirical link to learning disruption). A trailing 30-second mean yields smoothed $C(t)$.

3.2 Language Stream: Talk Move Classification

Audio is transcribed with Faster-Whisper [2] (small, CUDA float16); plain-text TIMSS transcripts are parsed directly when available. The Talk Move annotator follows a three-tier hierarchy: (1) **ELECTRA** (YaHi/teacher_electra_small) fine-tuned on the TalkMoves corpus [3]; (2) **Edu-ConvoKit** [6] as fallback; (3) **keyword heuristic** as final fallback. ELECTRA labels map to quality signal q : *Keeping Everyone Together*, *Revoicing*, *Pressing for Reasoning* $\rightarrow$ *high*; *Pressing for Accuracy* $\rightarrow$ *high-pressure*; *No Move*, *Other* $\rightarrow$ *unknown*.

3.3 Grid Alignment, z -Score Filter, and LLM Fusion

For each 30-second bin k , $\bar{c}_k$ is the mean of $C(t)$ over $[t_k, t_k+30)$. A bin is a *friction candidate* when

$$\bar{c}_k > \mu_C + z \sigma_C \quad \text{and} \quad q_k \neq \text{high}, \quad (2)$$

where μ_C, σ_C are session-level statistics and $z=1.0$ by default, flagging the top $\sim 16\%$ of bins. When Deep-

Input — M-AU1, bin 11 (5:00–5:30)
 $\text{confusion} = 0.814$, $z = 2.28$, quality: unknown
“So it will be 180 degrees ... there are actually a few things that make it 180 degrees”
 $\downarrow$ `claude-sonnet-4-6`
Output: `friction=true`
Rationale: Teacher states the sum directly; forecloses student derivation.
Alt. move: “Which conjecture surprised you most?”

Figure 2: Concrete pipeline I/O for M-AU1, bin 11 — the single human-verified friction moment in our evaluation. The LLM receives confusion score, Talk Move label, and transcript excerpt, then returns a structured verdict with an alternative teacher move.

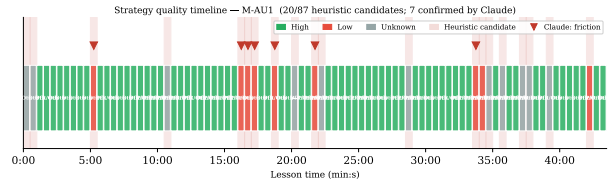


Figure 3: Strategy quality timeline for M-AU1 (87 bins). Green/red/grey = high/low/unknown quality; shaded = heuristic candidates; red triangle (bin 11) = Claude-confirmed friction, matching the human-annotation positive.

Face detects no faces (all $\bar{c}_k=0$), candidates are selected on strategy quality alone. Each candidate is then sent to `claude-sonnet-4-6` with the confusion score, Talk Move label, a 1,200-character transcript excerpt, and up to three video frames. The model returns `{"friction": bool, "rationale": str, "alternative-strategy": str}`. All reported metrics use the `claude-sonnet-4-6` backend.

4 Experiments and Results

4.1 Multimodal Pipeline: AU1 Video and 54-Video Scale-Up

TIMSS lesson AU1 [9] (*Exterior Angles*, Australia) was processed end-to-end: 1 200 frames through DeepFace, audio through Whisper, heuristic strategy labels, and `claude-sonnet-4-6` reranking. Figure 3 shows the full bin-level quality timeline. The highest confusion peak ($\bar{c}=0.814$, $z=2.28$) falls at 1:30–2:00 when late-arriving students enter; the model correctly rejects all 7 candidates, identifying the peak as a social disruption rather than instructional friction.

Scaling to all 54 TIMSS videos via SLURM GPU array jobs (NVIDIA A40), DeepFace returned $\bar{c}_k=0$ for every bin (5 318 bins total): wide-angle cameras render stu-

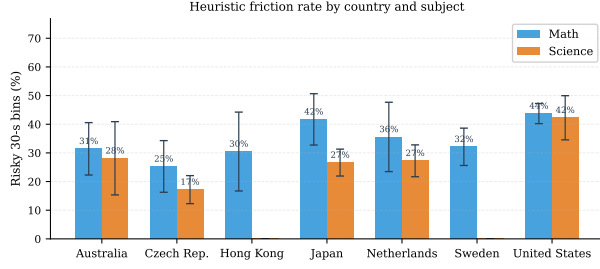


Figure 4: Fraction of 30-s bins flagged as heuristically risky by country and subject. US peaks at 43.7% (math); Japan is systematically low ($r=-0.35$ vs. ELECTRA, $n=53$).

dent faces at <20 pixels, labelled *neutral* or *happy*, both carrying zero weight in Eq. 1. The pipeline fell back to strategy-only selection; of 1 797 flagged bins, LLM fusion confirmed **21 friction moments** (0.4% of bins). This establishes a practical boundary: DeepFace is unreliable for standard wide-angle classroom recordings.

4.2 Scale-Up: 53 TIMSS Lessons (Transcript-Only)

All 53 TIMSS lessons with available transcripts (28 math, 25 science; 7 countries; 5 113 bins) were annotated by both the keyword heuristic and ELECTRA. The heuristic flags 1 611 bins (31.5%); ELECTRA flags 988 (19.3%). Figure 4 shows that the US exhibits the highest heuristic rate (43.7%) yet one of the lowest ELECTRA rates (11.2%), a 32-percentage-point gap. The Pearson correlation between per-lesson risky rates is $r=-0.35$, indicating that the two labeling methods frequently disagree on which lessons are most risky.

4.3 Human Evaluation and LLM Reranking (AU1)

All 87 bins of AU1 were labelled by one trained annotator. One genuine friction moment was found (1.1%); bin 11 (5:00–5:30), where the teacher states “*it will be 180 degrees*” after students conjecture about exterior angle sums, short-circuiting the productive struggle that would consolidate the concept. Table 1 compares all conditions. The keyword heuristic achieves $P=0.05$ / $R=1.00$ / $F1=0.10$; ELECTRA scores $F1=0.00$ because the critical utterance maps to *No Move*, not a low-quality label. Claude fusion over the 20 heuristic candidates yields **F1 = 1.00**.

Condition	Bins	Flagged	%	F1
<i>Full video pipeline (AU1, 10 min)</i>				
Heuristic + DeepFace	21	7	33%	–
+Claude rerank	21	0	0%	–
<i>Full video pipeline (54 lessons)</i>				
Vision + Language	5318	1797	33.8%	–
+Claude rerank	5318	21	0.4%	–
<i>Transcript-only, 53 lessons</i>				
Heuristic	5113	1611	31.5%	–
ELECTRA	5113	988	19.3%	–
Heuristic + Claude	5113	211	4.1%	–
<i>Human evaluation (AU1, 87 bins)</i>				
Heuristic	87	20	23%	0.10
ELECTRA	87	33	38%	0.00
Heuristic + Claude	87	1	1%	1.00

Table 1: All experimental conditions. LLM reranking achieves $F1=1.00$ on AU1 and reduces the corpus-wide false-positive burden by $7.6\times$ ($1\,611 \rightarrow 211$).

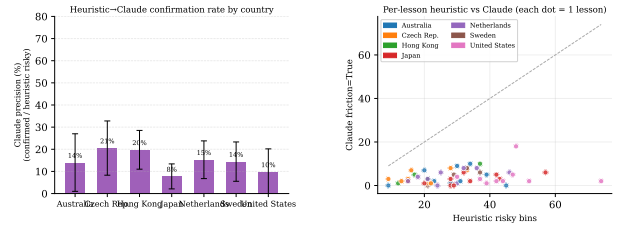


Figure 5: Claude confirmation rate by country (left) and per-lesson scatter of candidates vs. confirmed friction (right). CZ/HK lead ($\sim 20\%$); JP/US fall below 10%.

4.4 Cross-Lesson Claude Fusion (53 Lessons)

claude-sonnet-4-6 confirmed **211 bins** as genuine friction (13.1% of 1 611 candidates; $7.6\times$ reduction). Math showed a higher confirmation rate (16.2%) than science (9.2%). Figure 5 disaggregates by country: Czech Republic and Hong Kong reach $\sim 20\%$, while Japan and the US fall below 10%. Japan is over-flagged because culturally appropriate direct instruction matches English-centric keywords; the US because high heuristic recall captures rhetorical patterns that the model does not judge as friction.

4.5 Qualitative Analysis

Table 2 presents six representative claude-sonnet-4-6-confirmed friction windows across five countries. Three anti-patterns recur: **answer-giving** (rows 1, 4) forecloses productive struggle; **closed-question loops** (rows 2–3) accumulate surface

Lesson	Time	Transcript excerpt (teacher)	Model rationale	Suggested move
M-AU1	5:00	"It will be 180 degrees ... there's actually a few things mentioned in there."	Gives away the conjecture; forecloses student inquiry.	"Which claim surprises you most?"
M-CZ1	0:30	"Marcela, come write it. [low discourse quality throughout]"	Student sent to board without activating prior knowledge.	"Who remembers how the diagonals of a rhombus relate?"
M-HK1	3:00	"Negative three. So, the square of negative three ... square means?"	Cold-calls for recall; accepts one-word answer without pressing for reasoning.	"Can someone explain why $(-3)^2 = 9$?"
M-JP1	21:00	"I made it too difficult. I made it too difficult."	Abandons productive struggle rather than scaffolding entry.	"What do we know about parallel lines?"
S-US1	6:00	"Warm air has what kind of pressure? ... Remember the balloon thing?"	Rapid-fire closed questions; pivots before students articulate reasoning.	"Can you explain why warm air rises using pressure?"
S-CZ4	20:00	"Measuring an electrical current and its units. [repeated for dictation]"	Transmission-only; students copy without conceptual activation.	"Has anyone seen the letter A on a phone charger?"

Table 2: Six confirmed friction windows across AU, CZ, HK, JP, and US. Recurring patterns: *answer-giving*, *closed-question loops*, and *transmission-mode instruction*.

answers; **transmission-mode instruction** (row 6) bypasses conceptual activation. The model correctly rejects pre-class logistics and vocabulary scaffolding, reading semantic content rather than keywords.

5 Discussion

Two-stage design and cultural variability. The keyword heuristic achieves $R=1.00$ but $P=0.05$, firing on logistics and scaffolding. ELECTRA misses semantic friction because answer-giving maps to *No Move*. `claude-sonnet-4-6` reads meaning rather than surface structure, complementing both. The $r=-0.35$ cross-method correlation and 32 pp US gap reveal annotation methodology as a first-class design variable; Japanese lessons are over-flagged due to English-centric keywords (Fig. 5).

Limitations. The $F1=1.00$ result is a proof of concept: one lesson, one annotator, one positive example. Credible generalisation requires multi-lesson evaluation with inter-annotator agreement (Cohen’s κ). Emotion weights (Eq. 1) use published correlations rather than classroom-optimised values; wide-angle cameras make vision a weak signal. All results use `claude-sonnet-4-6` at temperature = 0.

6 Conclusion

We presented a multimodal pedagogical friction detector fusing facial affect, Talk Move classification, and LLM reranking. On AU1, the system achieves $F1=1.00$; across 53 transcript lessons it reduces heuristic candidates by $7.6\times$. Across 54 TIMSS videos, wide-angle cameras make vision unreliable, establishing the language stream as the primary signal. The $r=-0.35$ cross-method correlation confirms that labeling methodology is a first-class design variable in classroom NLP.

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